

Ethan Daniel Hershey

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EDUCATION

Penn State University, Main Campus

June 2022 – May 2026

Bachelor's Degree in Computer Engineering

State College, PA

- Minor: Mathematics (Science)
- Coursework: Data Structures/Algorithms, Computer Architecture, Operating Systems, Digital Logic, Networks, Signals/Systems, Differential Equations, SQL, Distributed Systems, OOP

TECHNICAL SKILLS

Programming Languages: Python, Rust, Java, C, C++, JavaScript, TypeScript, Go, Kotlin, Swift, HTML/CSS, SQL, MATLAB

Web Development: React, Axum (Rust framework), RESTful APIs, Node.js, Express, GraphQL

Cloud & DevOps: AWS (EC2, S3, Lambda), Git, Docker, Kubernetes, Linux (Fedora, Ubuntu, RHEL)

Software Engineering: Object-Oriented Design, Distributed Systems, Agile Development, Microservices

Machine Learning: TensorFlow, PyTorch, scikit-learn, Keras, NLTK, Hugging Face, Neural Networks, NLP

Hardware: Raspberry Pi, Arduino, ESP32, STM32, Microcontrollers

Other: 3D Modeling (Blender), Game Development (Bevy, GameMaker, Unity)

WORK EXPERIENCE

Decision Sciences & AI Intern

June 2025 – August 2025

CMI Media Group

Philadelphia, PA / Chesterbrook, PA

- Developed statistical test/control group creation tool using t-tests to maintain group similarities across selected metrics, reducing processing from 16 hours to 15 minutes (64x improvement)
- Analyzed 10TB+ of pharmaceutical health data using Python (pandas/polars) for hundred billion-dollar clients including Eli Lilly and Company
- Built text-to-SQL AI system using LangChain that democratized PostgreSQL database access for non-technical teams through natural language processing
- Delivered data-driven insights that directly influenced multi-million dollar marketing budget allocations across email campaigns, Google ads, and digital channels

Teaching Assistant for Computer Science 131

2022 – 2023

Penn State University

State College, PA

- Instructed 50+ students in object-oriented programming, data structures, and algorithm design
- Conducted code reviews and provided technical feedback while developing supplemental learning materials

Pool Manager & Lifeguard

2018 – 2023

Woodridge Swim Club

Lancaster, PA

- Managed operations and supervised 10+ lifeguards while implementing efficient scheduling system

PROJECTS & ACHIEVEMENTS

Multi-Server AI Movie Rating System (HackPSU Hackathon)

Fall 2023, Spring 2023, Fall 2024

- Built distributed system achieving 85% accuracy in predicting movie ratings from review sentiment using custom-trained NLP model with sub-200ms response time
- Collaborated with team of 4 to implement Python/GPU backend and React frontend in 24-hour hackathon

Full-Stack Personal Website

2023 – Present

- Deployed Axum (Rust) backend and React frontend on AWS EC2 with CI/CD pipeline for automated deployment

Video Game Development Portfolio

2022 – Present

- Developed 2D platformer in Bevy (Rust) with custom physics engine and procedural level generation

Smart Rain Water Management System

2022

- Won 1st Place at North Museum Science Fair for Raspberry Pi sensor network that reduced water waste using weather API integration and predictive modeling